Near-Wall Ultra-precise Motion Control of a Magnetically Driven Scanning Microprobe in Aqueous Solutions

Ta-Min Meng, Fei Long, *Senior Member, IEEE*, and Chia-Hsiang Menq

***Abstract*—As an untethered scanning probe approaches a wall in aqueous solutions, the drag coefficient increases and approaches infinity, which challenges the precise motion control of the scanning probe near the wall. This article presents a real-time estimation of space-dependent drag coefficient and unknown disturbances for a magnetically driven scanning microprobe close to a wall in aqueous solutions. The estimated drag coefficient is used to update the motion gain with a time-varying discrete-time control law to achieve a prespecified control objective for ultraprecise motion control of the scanning probe near the wall. The control law and the estimator have been integrated into a previously developed visual servo control system to achieve ultraprecise motion control of the scanning probe close to wall surfaces. The motion control system has been realized using a high-speed field programmable gate array (FPGA) system. Experiments have confirmed that near-wall ultraprecise motion control of the scanning probe has been achieved. In experiments, the tracking error of the microprobe was close to a random error with zero mean due only to random thermal forces and measurement noise. In addition, the accuracy of real-time estimation of motion gain/drag coefficient was experimentally verified.**

***Index Terms*—Electromagnetic actuation, estimation, motion control, visual servoing.**

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I. Introduction

MULTIPOLE electromagnetic actuators are being used in biological and medical applications involving the untethered manipulation of small objects, including the propulsion of miniatured swimming machines [1], [2], [3], [4], [5], [6], multiaxis atomic force microscopy (AFM) technology [7], [8], and the control of medical devices [9], [10]. Millimeter- and micrometer-scale swimming robots propelled in liquids using external magnetic fields have been successfully developed [1], [2], [3], [4], [5], [6]. The swimming motion of an untethered artificial bacterial flagella was demonstrated using a low-intensity rotating magnetic field [3]. Stable positioning and steering of microscopic magnetic beads in a 3-D workspace were achieved using a six-pole magnetic actuator and a visual servo control system [5]. A five-degree-of-freedom (DOF) magnetic actuator was designed to control intraocular microrobots for delicate retinal surgery [9], [10]. When controlling a microrobot in a visual servo framework, the error was less than 0.5 mm [10]. To expand the potential of microrobot applications, a haptic device was integrated into a magnetic tweezer via a network-based telemanipulation technique, which enabled remote manipulation of microrobots with haptic feedback [11].

Several multipole electromagnetic actuators have recently been developed to generate 2-D or 3-D magnetic forces [12], [13], [14], [15], [16]. Specifically, the hexapole actuator reported in [14] employs six pointed poles and uses six input currents to establish a controllable magnetic field in the 3-D workspace, where microprobes and samples are placed. A vision-based 3-D particle tracking system [17], [18] was integrated into the hexapole actuator to form an integrated visual servo control system to manipulate a selected microprobe as a scanning probe in aqueous solution for biological applications [19]. One of the core functions of this integrated system is the law of discrete-time motion control, which is designed according to the dynamics of microprobe motion in aqueous solution [20] to realize ultraprecise high-speed motion control of the microprobe.

However, when the probe is brought close to the sample, the drag coefficient increases significantly according to its distance from the sample surface. The local drag coefficient can be accurately measured from the variance of the Brownian motion of the microprobe [21]. This can be achieved by real-time estimation or experimental calibration before running the experiment. The work reported in [19] adopted the latter approach to model the drag coefficient of the microprobe near a solid plane [22].

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While the experiments reported in [19] have verified that excellent motion control has been achieved using the proposed motion control law, the reliance on the calibrated drag force model is critical for the probing system used in practical applications. If the sample geometry and mechanical properties are not known in advance, severe limitations may be imposed.

This article presents real-time estimation of space-dependent drag coefficient and unknown disturbances for a magnetically driven scanning microprobe close to a wall in aqueous solutions. The estimator has been integrated into a previously developed visual servo control system [19] and brings two novel features to the system. First, the estimated drag coefficient is used to update the motion gain with a time-varying discrete-time control law to achieve a prespecified control objective for ultraprecise motion control of the scanning probe near the wall. Second, the ability to estimate the space-dependent drag coefficient and unknown external forces in real time is an innovation necessary to maneuver the scanning probe in a working environment with objects with unknown positions and/or mechanical properties.

The content of this article is organized as follows. Section II describes the visual servo control of a scanning probe in aqueous solutions. The control system consists of five main functions: 1) ultraprecise visual tracking of 3-D motion; 2) Hall-sensor-based magnetic flux control; 3) real-time optimal flux allocations; 4) real-time updates of discrete-time motion control law; and 5) real-time estimation of motion gains and disturbances. This article mainly focuses on the fourth and fifth functions. Section III presents the experimental verification of real-time estimation of motion gain/drag coefficient and the experimental verification of ultraprecise and high-speed motion control of the scanning probe close to the wall. Finally, conclusions are drawn in Section IV.

II. Motion Control of a Scanning Probe in Aqueous Solutions

A previously developed experimental setup is shown in Fig. 1. It uses a six-input hexapole electromagnetic actuator to control the magnetic gradient force exerted on the selected microprobe to render untethered manipulation in aqueous solutions. The left panel shows the hexapole actuator integrated with an ultraprecise 3-D motion tracking system [18] to form a 3-D visual servo control system [19]. The design and fabrication of the hexapole actuator can be found in Section II-A of [14]. Specifically, the CAD model of the actuator is shown in Fig. 1, and the prototype is shown in Fig. 2 in [14].

The hexapole electromagnetic actuator uses six pointed rods, each made of ferrous steel, driven by individual coils. They are arranged in three pairs and placed symmetrically on three orthogonal axes, outlining the three drive axes and forming a hexagonal configuration with their tips enclosing the workspace where samples and the microprobe are placed. The workspace is a polyhedron whose size is determined by the distance between each pole tip and the center, denoted by ℓ. Although ℓ is adjustable, it was 500 μm for all experiments reported in this article.

Fig. 2 shows a block diagram of the control system designed to achieve ultraprecise 3-D visual servo control of the microprobe in aqueous solutions. It consists of five main functions:

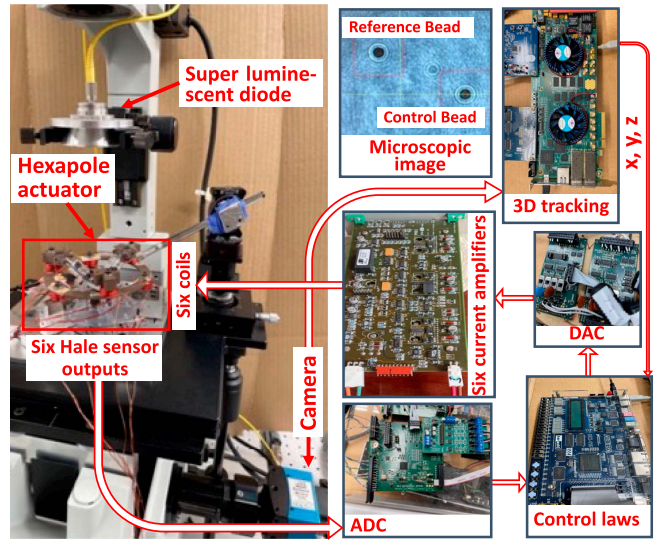

Fig. 1. Experimental setup for visual servo control of a magnetically driven scanning probe in aqueous solutions.

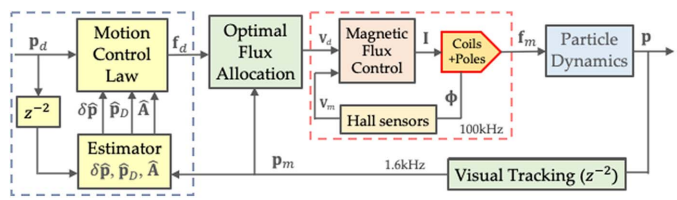

Fig. 2. Block diagram of the visual servo control system for untethered manipulation of a scanning microprobe in aqueous solutions.

1) ultraprecise visual tracking of 3-D motion; 2) Hall-sensor-based magnetic flux control; 3) real-time optimal flux allocations; 4) real-time updates of discrete-time motion control law; and 5) real-time estimation of motion gains and disturbances. The last four functions were implemented using an FPGA board (Altera: DE2-115), shown in the lower right panel. The six Hale sensor outputs are sent to the FPGA board through AD converters. The control effort is calculated and sent via DA converters to six current amplifiers. This article mainly focuses on the fourth and fifth functions but first briefly summarizes the most important results related to the first three functions.

A. Visual Tracking of 3-D Motion

The motion tracking system uses an inverted microscope (Olympus IX81) equipped with a high-speed CMOS camera (Mikrotron: MC3010) for image acquisition and an FPGA board (Terasic: Arria® V GT) for real-time calculation. The camera acquires 128×128-pixel images at frame rates up to 10 000 frames per second (fps) under a super luminescent diode (SLD) illumination system.

The ability to track 3-D motion is attributed to the use of microscope defocus images, i.e., the image of the spherical probe is related to its z-position, and this relationship is established by calibration before running the experiment. The live 2-D image of the probe is first compared with a calibrated object-specific image model using an optimal matching technique to estimate the axial position [23]. The transverse position

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of the probe is then determined using a continuous image registration method [24] to remove bias due to image asymmetry and the use of radial projections in previous implementations [17]. Three important objectives are achieved: 1) the real-time estimation of the 3-D position matches the maximum frame rate of the camera; 2) measurement bias is eliminated; and 3) the timing of the output data stream of the system is precisely controlled.

The top middle panel in Fig. 1 shows a snapshot of a simultaneous measurement of a moving probe and a fixed reference bead, where the two regions of interest are processed in real time to calculate the relative motion between the two beads. This dual bead tracking compensates for drift and eliminates low-frequency vibrations. Since it requires a larger region of interest (ROI), i.e., 512×512 pixels, the frame rate for visual servoing is 1.6 kHz. It has been verified by experiments that when using a $60\times$ magnification lens to measure a spherical probe with a radius of $2.25\,\mu$m, the resolution is about 0.7 nm (rms) in the x- and y-direction, and about 2.3 nm (rms) in the z-direction [18]. The total time delay attributed to image acquisition, data transmission, and real-time computation is two steps of the sampling cycle.

B. Hall-Sensor-Based Magnetic Flux Control

The hexapole actuator controls six input currents to generate magnetic fluxes at six-pole tips to establish a magnetic field in the 3-D workspace and control the 3-D magnetic gradient force exerted on the microprobe. A Hall-sensor-based six-input-six-output flux control system was designed and implemented to solve three problems in flux generation: 1) multipole coupling; 2) uncertainty caused by remanence and hysteresis; and 3) limited bandwidth. Experiments have shown that despite three problems with flux generation, using the improved six-input-six-output flux control, the flux at each pole tip can be precisely controlled individually at frequencies above 4 kHz without any experimental tuning [25]. More importantly, the experimental results fully agree with the theoretical predictions based on the design.

C. Real-Time Magnetic Flux Allocation

The hexapole actuator is an over-actuated strongly coupled nonlinear system with six input currents and three output force components. Together with the Hall-sensor-based flux control, a Hall-sensor-based force model was derived, characterizing the nonlinearity, coupling, position dependence, and redundancy of hexapole electromagnetic actuation. Whereas the force model can be used to predict the position-dependent 3-D magnetic force experienced by the microprobe in the workspace, the goal of using the actuator is to generate the required forces in practical applications.

The optimal flux allocation proposed in [26] and [27] was integrated with six-input-six-output flux control to generate precise magnetic forces. Fig. 3 shows two experimental results to verify the performance of magnetic force generation [25], where the required force along the x-axis in the first experiment was 5 pN, and the magnitude of the required force was increased to 10 pN in the second experiment. Similar results were also obtained when the *y-axis* or *z-axis* was selected in the experiment. The results show that combined with the optimized flux allocation, the developed flux control system can generate precise magnetic forces, enabling the six-input hexapole actuator to behave like a decoupled three-axis force generator with a bandwidth exceeding 4 kHz.

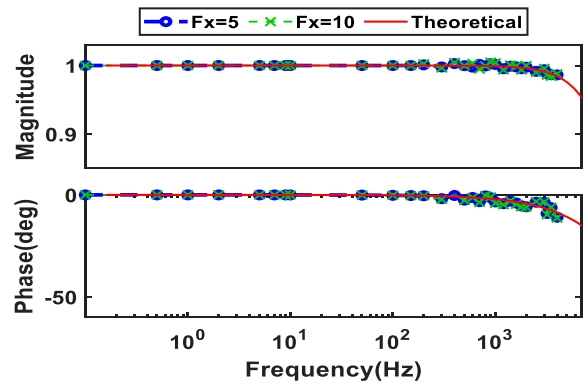

Fig. 3. Compares the frequency response of the three-axis magnetic force generation with the theoretical predictions based on the design.

D. Real-Time Updates of Discrete-Time Motion Control Law

The dynamics of a microscopic probe moving in an aqueous solution is governed by the Langevin equation at low-Reynolds numbers [19], driven by controllable magnetic forces $\mathbf{f}_m$ and influenced by unknown disturbances $\mathbf{f}_D$ and random thermal forces $\mathbf{f}_T$. The equation for 3-D motion is expressed as

$$\mathbf{\Gamma}(\mathbf{p}(t))\dot{\mathbf{p}}(t) = \mathbf{f}_m(\mathbf{p}(t), \boldsymbol{\phi}(t)) + \mathbf{f}_D(t) + \mathbf{f}_T(t) \tag{1}$$

where $\mathbf{\Gamma}$ is the 3×3 drag coefficient matrix and $\mathbf{p}$ is the position of the probe in the 3-D workspace. The magnetic force is position-dependent and controlled by $\boldsymbol{\phi}$, which is the flux vector at the six pole tips of the hexapole actuator. In practice, the flux vector is linearly related to the voltage readings of the Hall sensors by $\boldsymbol{\phi} = \mathbf{D}_H \mathbf{v}_H$, where $\mathbf{D}_H$ is a diagonal voltage-flux gain matrix and $\mathbf{v}_H$ is the sensor-voltage vector [27]. The Jacobian matrix derived from the position-dependent force model, $\mathbf{J} = \partial \mathbf{f}_m / \partial \mathbf{p}$, can be used to verify the instability of the dynamic system in (1) in the absence of motion feedback control.

The nominal value of the drag coefficient is characterized by Stokes' law for spherical probes, i.e., $\gamma_N = 6\pi\eta R$, where η is the dynamic viscosity of the solution and R is the probe's radius. As the probe approaches a wall, the drag coefficient varies with its distance from the surface. A dimensionless variable h/R, where h is the distance from the center of the probe to the wall, and two principal directions, perpendicular and parallel to the wall, characterize the space-dependent drag coefficient, i.e., $\gamma_\perp = \gamma_N \cdot C_\perp(h/R)$ and $\gamma_\parallel = \gamma_N \cdot C_\parallel(h/R)$. If the wall is a rigid plane, $C_\perp(h/R)$ and $C_\parallel(h/R)$ are two known space-dependent correction functions [21]. However, depending on the geometry (e.g., curved surface) and the mechanical properties of the wall (e.g., living cells), their analytical formula may not exist. In addition, when the probe approaches the surface, h/R tends to 1, and the function values of $C_\perp(h/R)$ and $C_\parallel(h/R)$ grow rapidly and approach infinity, which poses a challenge to the precise motion control of the scanning probe near the wall.

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The wall is the top surface of the petri dish in the experiment, which is flush with the xy plane of the measurement coordinate system. Therefore, the drag coefficient matrix is now a depth-dependent diagonal matrix, i.e., $\mathbf{\Gamma}(\mathbf{p}) = diag(\gamma_x(z), \gamma_y(z), \gamma_z(z))$. The depth-dependent drag coefficient $\gamma(z)$ can be accurately measured from the variance of the Brownian motion of the microprobe [21]. This can be achieved by real-time estimation or experimental calibration before running the experiment. The latter approach was employed in [19] to achieve precise motion control of a microprobe in aqueous solutions. In this article, a real-time estimator is designed to estimate the drag coefficient and update the motion gain with a discrete-time control law for near-wall ultraprecise motion control of a magnetically driven microprobe in aqueous solutions.

The zeroth-order hold is used to realize digital control. Although the control input remains constant within a sampling interval, the probe continues to move, and the magnetic force varies with the position of the probe. For $k\Delta t \le t < (k+1)\Delta t$, the first-order Taylor expansion is used to approximate the position-dependent magnetic force

$$\mathbf{f}_m(\mathbf{p}(t), \boldsymbol{\phi}(t)) \approx \mathbf{f}_m(k) + \mathbf{J}(k)\delta\mathbf{p}(t) \tag{2}$$

where $\mathbf{f}_m[k] = \mathbf{f}_m(\mathbf{p}[k], \boldsymbol{\phi}[k])$, $\mathbf{J}[k] = \partial\mathbf{f}_m/\partial\mathbf{p}(\mathbf{p}(k), \boldsymbol{\phi}(k))$, and $\delta\mathbf{p}(t) = \mathbf{p}(t) - \mathbf{p}[k]$. It was validated that by combining optimal flux allocation with six-input-six-output flux control, the six-input hexapole actuator behaves like a decoupled three-axis force generator with a bandwidth exceeding 4 kHz. It can generate the magnetic force $\mathbf{f}_m[k]$ in (2) to precisely follow $\mathbf{f}_d[k]$, which is the desired force to be determined from the motion control law. A time-varying equation of motion can be derived from (1) to describe the motion of the controlled microprobe over the sampling interval, i.e., $k\Delta t \le t < (k+1)\Delta t$,

$$\begin{aligned}\delta\dot{\mathbf{p}}(t) &= \mathbf{\Gamma}^{-1}(\mathbf{p}(t))\mathbf{J}[k]\delta\mathbf{p}(t)\\ &\quad + \mathbf{\Gamma}^{-1}(\mathbf{p}(t))\{\mathbf{f}_d[k] + \mathbf{f}_D(t) + \mathbf{f}_T(t)\}.\end{aligned} \tag{3}$$

The motion $\delta\mathbf{p}(\Delta t)$ within the sampling interval can be derived from (3). It can then be used to cast a time-varying discrete-time equation of motion for the digital control system

$$\mathbf{p}[k+1] = \mathbf{p}[k] + \mathbf{A}[k]\{\mathbf{f}_d[k] + \mathbf{f}_D[k] + \mathbf{f}_T[k]\} \tag{4}$$

where when $||\mathbf{J}[k]||$ is small, the gain matrix can be approximated as a diagonal matrix, i.e., $\mathbf{A}[k] \approx \mathbf{\Lambda}[k] = diag(\Delta t\gamma_x^{-1}[k], \Delta t\gamma_y^{-1}[k], \Delta t\gamma_z^{-1}[k])$. Since the 3-D motion of the controlled probe is decoupled, for simplicity, the equation for 1-D motion is given in the following analysis:

$$x[k+1] = x[k] + a_x[k]\left(f_{dx}[k] + f_T[k]\right) + x_D[k] \tag{5}$$

where $a_x[k] = \Delta t\gamma_x^{-1}[k]$ is the motion gain, f_{dx} is the desired force determined from the control law, f_T is the thermal force experienced by the probe, and $x_D[k] = a_x[k]f_{Dx}[k]$ is the disturbed motion attributed to $f_{Dx}[k]$, the combination of modeling error and disturbance.

With the control objective being $\delta x[k+1] = \lambda_c \cdot \delta x[k]$, where $|\lambda_c| < 1$, $\delta x[k] = x_d[k] - x[k]$, and $x_d[k]$ is the desired motion trajectory, the desired force f_{dx} can be derived by comparing (5) with the desired motion

$$f_{dx}[k] = \hat{a}_x^{-1}[k]\{x_d[k+1] - x_d[k] + (1-\lambda_c)\delta\hat{x}[k] - \hat{x}_D[k]\} \tag{6}$$

where $\hat{a}_x[k]$ is the estimate of motion gain, $\delta\hat{x}[k]$ is the estimate of tracking error, and $\hat{x}_D[k]$ is the estimate of disturbed motion $x_D[k]$. Substituting $f_{dx}[k]$ from the motion control law (6) into (5), the dynamics of tracking error can be derived by comparing the resulting motion with the desired motion trajectory

$$\delta x[k+1] = \lambda_c\delta x[k] - \varepsilon[k] \tag{7}$$

where $\varepsilon[k]$ is attributed to random thermal forces and three estimation errors: $e_{\delta x}[k] = \delta x[k] - \delta\hat{x}[k]$, $e_a[k] = a_x[k] - \hat{a}_x[k]$, and $e_{x_D}[k] = x_D[k] - \hat{x}_D[k]$

$$\varepsilon[k] = a_x[k]f_T[k] - (1-\lambda_c)e_{\delta x}[k] + e_a[k]f_{dx}[k] + e_{x_D}[k]. \tag{8}$$

It can be seen from (7) and (8) that if the unbiased estimation of $a_x[k]$, $\delta x[k]$, and $x_D[k]$ can be realized, then $\varepsilon[k]$ is a random variable with zero mean, and the tracking error is a random error with zero mean due only to random thermal forces and measurement noise.

E. Real-Time Calculation of Process Outputs

The measured tracking error consists of two components: 1) a deterministic component; and 2) a random component subject to random measurement noise and thermal forces. A single-pole infinite impulse response (IIR) filter [28] is used to estimate the deterministic component of the measured tracking error

$$\overline{\delta x}_m[k+1] = a_{\text{var}} \cdot \delta x_m[k] + (1 - a_{\text{var}})\overline{\delta x}_m[k] \tag{9}$$

where $(1 - a_{\text{var}})$ is the pole position, which determines the effective length over which the deterministic component of the measured tracking error is calculated. The random component is defined as $\delta x_r[k] = \delta x_m[k] - \overline{\delta x}_m[k]$. The same IIR filter is used to calculate the mean and the mean square of $\delta x_r[k]$. They are then used to calculate the variance of the random component, i.e.,

$$\sigma^2_{\delta x_r}[k] = \overline{\delta x_r^2}[k] - \left(\overline{\delta x}_r[k]\right)^2. \tag{10}$$

The thermal force experienced by the probe, f_T, is quantified by variance, which is proportional to γ and inversely proportional to Δt, i.e., $\sigma^2_{f_T} = 4k_BT\gamma/\Delta t$ [29], where k_B is the Boltzmann's constant and T is the absolute temperature of the aqueous solution. When the motion tracking system has a two-step measurement delay, the relationship between the variance of tracking error and that of thermal motion and of measurement noise was derived in [19]

$$\sigma^2_{\delta x} = \left(2 + \frac{1}{1-\lambda_c^2}\right)\sigma^2_{\delta x_T} + \left(\frac{1-\lambda_c}{1+\lambda_c}\right)\sigma^2_{n_x} \tag{11}$$

where $\sigma^2_{\delta x_T} = (\Delta t/\gamma)^2\sigma^2_{f_T} = 4k_BTa_x[k]$ is the variance of thermal motion and $\sigma^2_{n_x}$ is the variance of measurement noise. Since the measured tracking error is used to calculate the variance of the

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random component, an additional measurement noise needs to be added to (11)

$$\sigma^2_{\delta x_r}[k] = \left(2 + \frac{1}{1-\lambda_c^2}\right) 4k_b T a_x[k] + \left(\frac{2}{1+\lambda_c}\right)\sigma^2_{n_x}. \tag{12}$$

Thus, in addition to $\delta x[k]$, a second output of the controlled system, which is the measured input gain, can be derived from (12)

$$a_{xm}[k] = \left(\sigma^2_{\delta x_r}[k] - \left(\frac{2}{1+\lambda_c}\right)\sigma^2_{n_x}\right) \Big/ \left(4k_b T\left(2 + \frac{1}{1-\lambda_c^2}\right)\right) \tag{13}$$

which is an additional measurement output. Therefore, the measurement at time k is $\boldsymbol{y}[k] = \begin{bmatrix}\overline{\delta x}_m[k] & a_{xm}[k]\end{bmatrix}^T$, the two components of which are calculated from $\delta x_m[k]$ in real time and are to be used by the augmented estimator discussed as follows.

F. Real-Time Estimation of Motion Gain, Motion Error, and Disturbance

Real-time estimation of $\hat{a}_x[k]$, $\delta\hat{x}[k]$, and $\hat{x}_D[k]$ is required to update the discrete-time control law (6). Due to the two-step measurement delay, it is necessary to use three variables, $\delta x_1[k] = \delta x[k-2]$, $\delta x_2[k] = \delta x[k-1]$, and $\delta x_3[k] = \delta x[k]$, to model the dynamics of motion error. Two separate second-order processes are employed to model the disturbed motion $x_D[k]$ attributed to $f_{Dx}[k]$ and the time-varying motion gain $a_x[k]$, respectively. Therefore, seven variables are used to depict the true state of the process at time $(k+1)$ evolved from the state at k

$$\begin{cases} \delta x_1[k+1] = \delta x_2[k] \\ \delta x_2[k+1] = \delta x_3[k] \\ \delta x_3[k+1] = \lambda_c \delta x_3[k] + (1-\lambda_c)e_{\delta x}[k] - e_{x_D}[k] \\ \qquad\qquad - f_{dx}[k]e_a[k] - a_x[k]f_T[k] \\ x_D[k+1] = x_D[k] + \delta x_D[k] \\ \delta x_D[k+1] = \delta x_D[k] \\ a_x[k+1] = a_x[k] + \delta a_x[k] \\ \delta a_x[k+1] = \delta a_x[k] \end{cases} \tag{14}$$

where the third equation is derived from (7) and (8), and the process noise is $a_x[k]f_T[k]$.

The measurement at time k is $\boldsymbol{y}[k] = \begin{bmatrix}\overline{\delta x}_m[k] & a_{xm}[k]\end{bmatrix}^T$, the two components of which are to be compared with the output of the estimator, $\hat{\boldsymbol{y}}[k] = \begin{bmatrix}\delta\hat{x}_1[k] & \hat{a}_x[k]\end{bmatrix}^T$. Thus, the output matrix of the seven-variable process is cast as a 2×7 matrix

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}. \tag{15}$$

Through the feedback of two estimation errors $e_{x_1}[k] = \overline{\delta x}_m[k] - \delta\hat{x}_1[k]$ and $e_{a_x}[k] = a_{xm}[k] - \hat{a}_x[k]$, the closed-loop estimator is designed as

$$\begin{cases} \delta\hat{x}_1[k+1] = \delta\hat{x}_2[k] + \{\ell_{11}[k]e_{x_1}[k] + \ell_{12}[k]e_{a_x}[k]\} \\ \delta\hat{x}_2[k+1] = \delta\hat{x}_3[k] + \{\ell_{21}[k]e_{x_1}[k] + \ell_{22}[k]e_{a_x}[k]\} \\ \delta\hat{x}_3[k+1] = \lambda_c\delta\hat{x}_3[k] + \{\ell_{31}[k]e_{x_1}[k] + \ell_{32}[k]e_{a_x}[k]\} \\ \hat{x}_D[k+1] = \hat{x}_D[k] + \delta\hat{x}_D[k] \\ \qquad\qquad + \{\ell_{41}[k]e_{x_1}[k] + \ell_{42}[k]e_{a_x}[k]\} \\ \delta\hat{x}_D[k+1] = \delta\hat{x}_D[k] + \{\ell_{51}[k]e_{x_1}[k] + \ell_{52}[k]e_{a_x}[k]\} \\ \hat{a}_x[k+1] = \hat{a}_x[k] + \delta\hat{a}_x[k] \\ \qquad\qquad + \{\ell_{61}[k]e_{x_1}[k] + \ell_{62}[k]e_{a_x}[k]\} \\ \delta\hat{a}_x[k+1] = \delta\hat{a}_x[k] + \{\ell_{71}[k]e_{x_1}[k] + \ell_{72}[k]e_{a_x}[k]\} \end{cases} \tag{16}$$

where $\ell[k]$'s are entries of the 7×2 feedback matrix $\mathbf{L}[k]$ of the estimator. The dynamic of estimation errors $\mathbf{e}$ is derived by subtracting (16) from (14)

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\mathbf{e}[k] - \mathbf{L}[k]\mathbf{H}\mathbf{e}[k] + \mathbf{q}[k] \tag{17}$$

where $\mathbf{q}[k] = a_x[k]f_T[k] \cdot \begin{bmatrix}0 & 0 & -1 & 0 & 0 & 0 & 0\end{bmatrix}^T$ is the process noise and $\mathbf{F}_e[k]$ is the state transition matrix of the error dynamics of the estimator

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -f_{dx}[k] & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \tag{18}$$

The estimator feedback matrix $\mathbf{L}[k]$ is updated according to $\mathbf{F}_e[k]$ and $\mathbf{H}$ by the recursive calculation used in the Kalman filtering method [25], [28], [30]

$$\mathbf{L}[k] = \mathbf{P}^f[k]\mathbf{H}^T\left(\mathbf{H}\mathbf{P}^f[k]\mathbf{H}^T + \mathbf{R}[k]\right)^{-1} \tag{19}$$

where $\mathbf{P}^f[k]$ is the forecast of the error covariance matrix, and the covariance matrix $\mathbf{R}[k]$ of the measurement noise is derived from the variance of n_x. The estimator feedback matrix $\mathbf{L}[k]$ is then used in (16) to update the seven estimates and to update the error covariance matrix

$$\mathbf{P}[k] = (\mathbf{I} - \mathbf{L}[k]\mathbf{H})\mathbf{P}^f[k] \tag{20}$$

which is then projected to the next step to update the forecast of the error covariance matrix

$$\mathbf{P}^f[k+1] = \mathbf{F}_e[k]\mathbf{P}[k]\mathbf{F}_e^T[k] + \mathbf{Q}[k] \tag{21}$$

where the covariance matrix $\mathbf{Q}[k]$ of the process noise is related to the variance of thermal force according to $a_x[k]f_T[k] \approx \hat{a}_x[k]f_T[k]$. Since $e_{a_x}[k] = a_{xm}[k] - \hat{a}_x[k] \approx a_x[k-2] - \hat{a}_x[k]$, the estimated motion gain has a delay of two steps.

III. Experimental Verification of Ultraprecise Visual Servo Control

This section uses experimental results to demonstrate that near-wall ultraprecise motion control of a scanning microprobe

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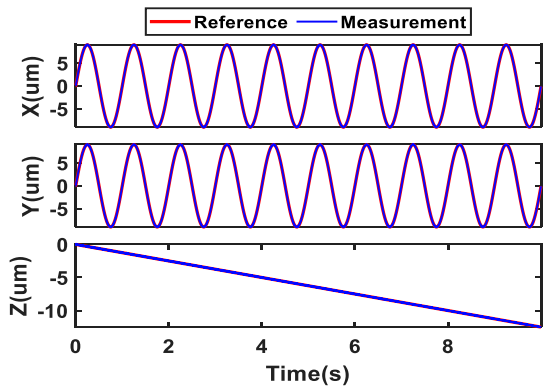

Fig. 4. Compare the three components of the 3-D motion of the precisely controlled microprobe with the desired reference trajectory.

in aqueous solutions can be achieved by updating the discrete-time control law with real-time estimates of process parameters. In all experiments, $\lambda_c = 0.4$ was the value used in the law of discrete-time motion control and the radius of the magnetic microprobe was 2.25 μm.

A. Ultraprecise Motion Control

Two experiments were performed to demonstrate the importance of real-time drag coefficient estimation for tracking performance. The microprobe started from the initial position $(0, 0, 0\,\mu\text{m})$. The desired z-motion of the probe was to move a 12.5 μm ramp in the negative z-direction, approaching the wall. The desired x- and y-motion were the same. They were 1 Hz sine waves with an amplitude of 9 μm. The entire controlled movement lasted 10 s. In the first experiment, the probe was stabilized and controlled by the time-varying discrete-time control law (6) with real-time estimation of process parameters. In the experiment, the wall was placed at $z = -15\,\mu$m and the microprobe was immersed in water. Since the ramp was 12.5 μm, the closest distance from the center of the probe to the wall was 2.5 μm. Since the probe radius was 2.25 μm, h/R tended to 1.111. Fig. 4 shows the three components of the resulting 3-D motion. It can be seen from the results that the microprobe was controlled to track the desired trajectory precisely.

The left panel of Fig. 5 displays the corresponding tracking error. It can be seen from the experimental results that the tracking error of the microprobe was close to a random error with zero mean, and its variance decreased when approaching the wall. In the second experiment, although the desired motion was the same, instead of the real-time estimated drag coefficient, the nominal drag coefficient γ_N was used in the control law. In addition, the disturbance compensation was turned off. The resulting tracking error is shown in the right panel of Fig. 5. It can be seen from the experimental results that the tracking error had two components: 1) deterministic component; and 2) random component, and the deterministic component gradually increased while approaching the wall. The increase in the tracking error was due to the increase in the difference between the true drag coefficient and the nominal drag coefficient.

The three motion gains of the 3-D probing system were estimated in real time in the experiment. They are displayed in

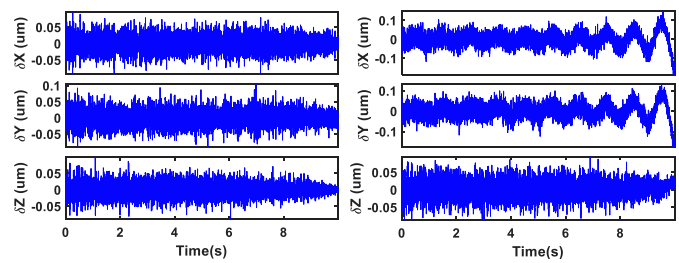

Fig. 5. Left: Tracking error by the proposed control law with a real-time estimation of process parameters. Right: Tracking error when the nominal drag coefficient was used in the control law and the disturbance compensation was turned off.

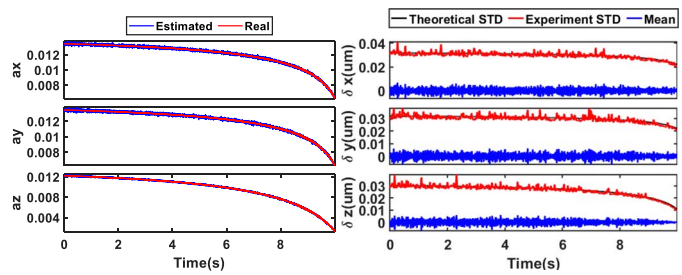

Fig. 6. Left: Compares the real-time estimated three motion gains with their true values. Right: The deterministic component and the standard deviation of the random component of the tracking error.

the left panel of Fig. 6. They are compared with the real values calculated using $\gamma_\perp = \gamma_N \cdot C_\perp(h/R)$ and $\gamma_\parallel = \gamma_N \cdot C_\parallel(h/R)$, where h is the distance from the center of the probe to the wall [22]. It can be seen from the results that the real-time estimated motion gains tracked closely the real values. With the real-time updated estimator feedback matrix $\mathbf{L}[k]$, the estimator minimizes the parameter error $\mathbf{e}[k]$ in (17), which subsequently minimizes the deterministic component of $\varepsilon[k]$ in (8). Therefore, the tracking error dynamics described by (7) is dictated by random thermal forces, i.e., $\varepsilon[k] \approx a_x[k] f_T[k]$.

The IIR filter in (9) was used to separate the mean and the random component of the measured tracking error, as shown in the left panel of Fig. 5. The mean is displayed in blue in the right panel of Fig. 6, wherein the mean of the tracking error was close to zero for all three axes. The standard deviation of the random tracking error was calculated and compared with the theoretical predictions in the right panel of Fig. 6. It can be seen from the comparison that the standard deviation of the random tracking error was in good agreement with theoretical predictions.

B. High-Speed Near-Wall-Surface Motion Control

It can be seen from the previous two experiments that it is very important to realize real-time estimation to realize the time-varying discrete-time control law when controlling the near-wall motion of the microprobe. Two experiments were conducted focusing on high-speed motion control close to the wall. Instead of slowly approaching the wall from a distance, the probe started from an initial position close to the wall $(0, 0, -11.5\,\mu\text{m})$, and the desired z-motion was a 3 Hz sine wave with an amplitude of 1 μm. Therefore, the closest distance

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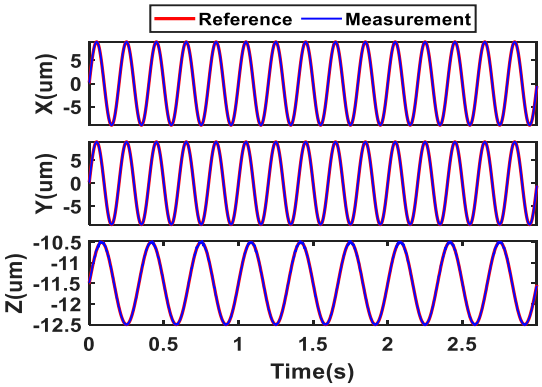

Fig. 7. Compare the three components of the 3-D motion of the precisely controlled microprobe with the desired reference trajectory close to the wall.

from the wall to the center of the probe center was 2.5 μm. The desired x- and y-motion were the same. They were 5 Hz sine waves with an amplitude of 9 μm. The entire controlled movement lasted 3 s. In the first experiment, the probe was stabilized and controlled by the time-varying discrete-time control law (6) with real-time estimation of process parameters. Fig. 7 shows the three components of the resulting 3-D motion. It can be seen from the results that the microprobe was controlled to track the desired trajectory precisely.

The left panel of Fig. 8 shows the corresponding tracking error in the first experiment. It can be seen from the experimental results that the tracking error of the microprobe was close to a random error with zero mean, and the variation of its variance was synchronous with the sinusoidal z-motion of the microprobe. The right panel of Fig. 8 shows the resulting tracking error in the second experiment, wherein the nominal drag coefficient γ_N was used in the control law and the disturbance compensation was turned off. The deterministic component of the tracking error was evident for all three axes. Specifically, the tracking errors for x- and y-motion fluctuated between $\pm 0.8\,\mu$m, while the error for z-motion fluctuated between $+0.12$ and $-0.18\,\mu$m.

The three motion gains were estimated in real time and are displayed in the left panel of Fig. 9. They are compared with the real values determined by the wall effect model [22]. It can be seen from the results that the estimated motion gains tracked closely the real values. In addition, the variation of motion gains was synchronous with the sinusoidal z-motion of the probe. The IIR filter in (9) was used to separate the mean and the random component of the measured tracking error, as shown in the left panel of Fig. 8. The mean of the tracking error was close to zero for all three axes, displayed in blue in the right panel of Fig. 9. The standard deviation of the random error was calculated and compared with the theoretical predictions in the right panel of Fig. 9. It can be seen from the comparison that the standard deviation was in good agreement with theoretical predictions.

C. 3-D Rapid Steering

A 3-D rapid steering experiment was reported in (18), in which a precalibrated drag coefficient model was used in the motion control law to achieve precise 3-D rapid steering of the probe. Instead of using a precalibrated model, the same

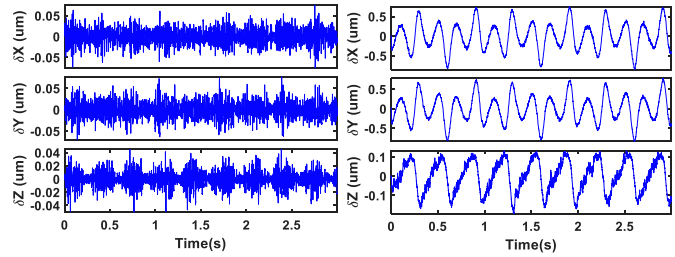

Fig. 8. Left: Tracking error by updating the discrete-time control law with real-time estimation. Right: Tracking error when the nominal drag coefficient was used and the disturbance compensation was turned off.

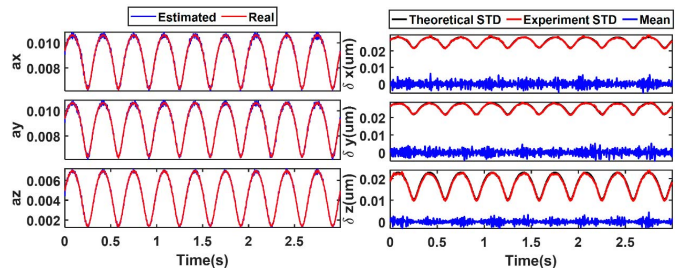

Fig. 9. Left: Compares the real-time estimated three motion gains with their true values. Right: The deterministic component and the standard deviation of the random component of the tracking error.

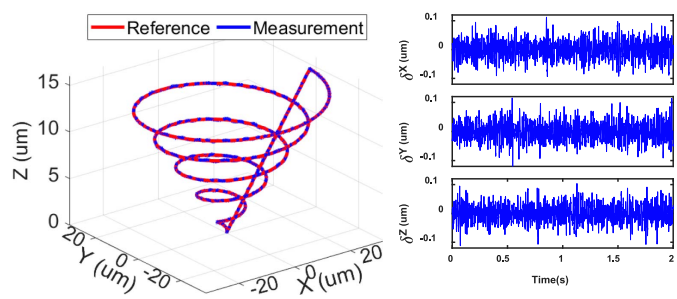

Fig. 10. 3-D rapid steering by updating the discrete-time control law with real-time estimation of process parameters. Left: Resulting 3-D motion; and right: three components of the tracking error.

experiment was performed by updating the discrete-time motion control law with real-time estimates of motion gains, tracking errors, and disturbed motions. In the experiment, the top surface of the petri dish was placed at $z = -15\,\mu$m, and the microprobe was immersed in water, the probe started from the initial position of $(0, 0, 0\,\mu\text{m})$, was controlled to rotate along the upward pitch for five cycles, each cycle taking 0.35 s, reached the highest point $(30, 0, 15\,\mu\text{m})$, and then was steered back to the initial position in 0.25 s.

Fig. 10 presents the experimental results of 3-D rapid steering in aqueous solutions. The left panel compares the resulting 3-D motion with the specified trajectory, and the right panel displays the tracking errors of the three axes. The results show ultraprecise rapid 3-D steering with no deterministic errors in any of the three axes. The three motion gains were estimated in real time to update the discrete-time control and are displayed in the left panel of Fig. 11. They are compared to the real values determined by the wall effect model [22]. It can be seen from the results that the real-time estimated motion gains tracked closely the true values.

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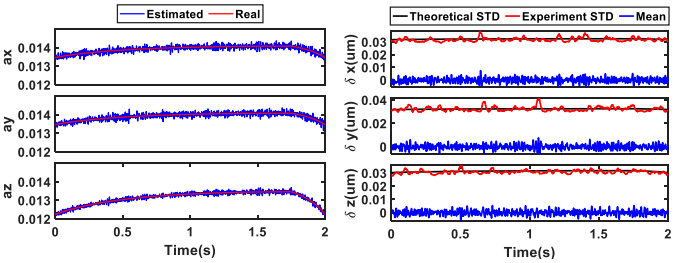

Fig. 11. Left: Compares the real-time estimated three motion gains with their true values. Right: The deterministic component (blue) and the standard deviation (red) of the random component of the tracking error.

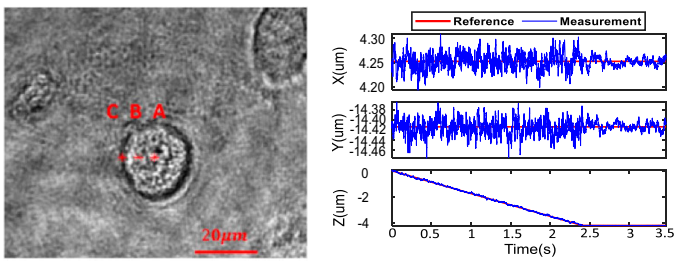

Fig. 12. Left: Light microscopy image snapshot of HeLa cells. Point A represents the specified single-point indentation location. Right: The measured 3-D motion of the probe (blue) compared to the reference trajectory (red), illustrating that the motion error was close to a random error with zero mean.

The mean and the random component of the measured tracking error were separated and shown in the right panel of Fig. 11, wherein the mean of the tracking error was close to zero for all three axes, and the standard deviation of the random error was in good agreement with theoretical predictions.

D. Near-Living Cell Motion Control and Vertical Indentation

Utilizing the ability to estimate process parameters and unknown disturbances in real time, precise control of the vertical indentation process of living cells is realized. This section presents the experimental results.

HeLa cells were seeded onto coverslips in a six-well cell culture plate. Cells were then cultured in an incubator (Forma Series II 3110, Thermo Scientific) at 37 °C and with 5% CO2 for 48 h. The coverslip with the best cell condition and density was selected for the experiment. Light microscopy imaging was used to find isolated cells and specify the desired indentation site. In the experiment, as shown in Fig. 12 (left), a round cell was selected, and the top center point A of the cell was selected as the desired indentation site.

Image quality of cells was degraded due to the use of a 60× objective for motion tracking, making image sharpness more sensitive to focal plane adjustments than typical light microscopy cell imaging. In the experiment, the focal plane was initially set above the cell of interest and then lowered to a position close to the top of the cell to estimate cell height. It was subsequently lowered until the image boundaries of the cells could be clearly observed. Since the goal was simply to find cells of different heights, no effort was made to find optimal image quality.

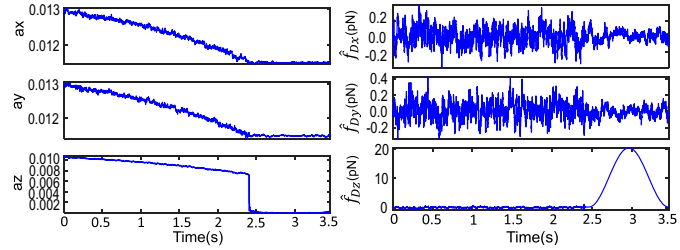

Fig. 13. Left: Three motion gains estimated in real time. Right: Three components of the prove/cell interaction force estimated in real time.

In addition, images were acquired by a high-speed CMOS camera at a frame rate of 1.6 kHz, further reducing image quality due to short exposure times.

In the experiment, the probe was controlled to move vertically downward at a constant speed of $1.6\,\mu\mathrm{m/s}$, approaching the indentation site of the cell, while the motion gain gradually decreased. Once the estimated motion gain reached 50% of the nominal value, while stabilizing the probe by motion control, a magnetic force of 0.25 pN was applied in the negative z-direction for 0.0625 s to ensure that the probe was in contact with the cell. The desired force $f_{\mathrm{intdz}}[k] = 10[1 - \cos(k\omega/\Delta t)]$ was subsequently applied to indent the cell. Fig. 12 (right) shows the measured 3-D motion of the probe compared to the reference trajectory. It can be seen from the experimental results that the motion error of the microprobe was close to a random error with zero mean, and the variance was significantly reduced when encountering the cell.

The estimated motion gain is shown in Fig. 13, left panel. It is seen that the estimated motion gain of the probe dropped sharply after encountering the cell. This is because the stiffness of the cell introduced significant resistance into the dynamics. The estimated interaction forces/disturbances are shown on the right panel. It is worth noting that the estimator does not directly estimate the disturbance experienced by the probe. Instead, the disturbance needs to be calculated using the estimated motion gain and the disturbed motion, e.g., $\hat{f}_{Dz} = \hat{z}_D/\hat{a}_z$. It can be seen from the experimental results that the estimated disturbance/interaction force accurately followed the expected force during indentation. During noncontact periods, the estimated disturbances were dominated by thermal forces.

IV. Conclusion and Discussion

Real-time estimation of space-dependent drag coefficient and unknown disturbances was designed, implemented, and integrated into a previously developed visual servo control system to achieve ultraprecise, high-speed, untethered manipulation of a magnetically driven scanning probe in aqueous solutions. It brings two novel features to the integrated system. First, the estimated drag coefficient can be used to update the motion gain with the time-varying discrete-time control law for ultraprecise motion control of the scanning probe near-wall surfaces. Second, the ability to estimate in real time the space-dependent drag coefficients and unknown external forces experienced by the scanning probe is necessary for maneuvering the scanning probe

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in working environments with objects with unknown positions and/or mechanical properties.

Experimental results demonstrated that ultraprecise motion control of a scanning microprobe was achieved by updating the discrete-time control law with real-time estimates of process parameters. Specifically, experimental results showed that the scanning probe could be controlled to precisely track the desired trajectory close to a wall whose surface normal is known. In addition, the tracking error of the scanning probe was close to a random error with zero mean, and the variation of its variance was in good agreement with the theoretical prediction.

Several aspects can be worked on to improve the functionality of the control system. The gain matrix can be approximated as a diagonal matrix to decouple the 3-D motion control when the wall normal is known. When applying the control system to scan the probe around 3-D samples whose 3-D geometry and mechanical properties are unknown, the ability to estimate the surface normal and space-dependent drag coefficients in real time is necessary. It lays the foundation for the probing system to realize automated scanning and probing of previously unknown or unspecified samples in the workspace.

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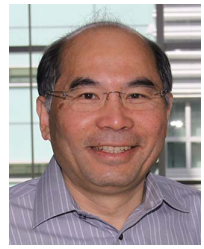

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